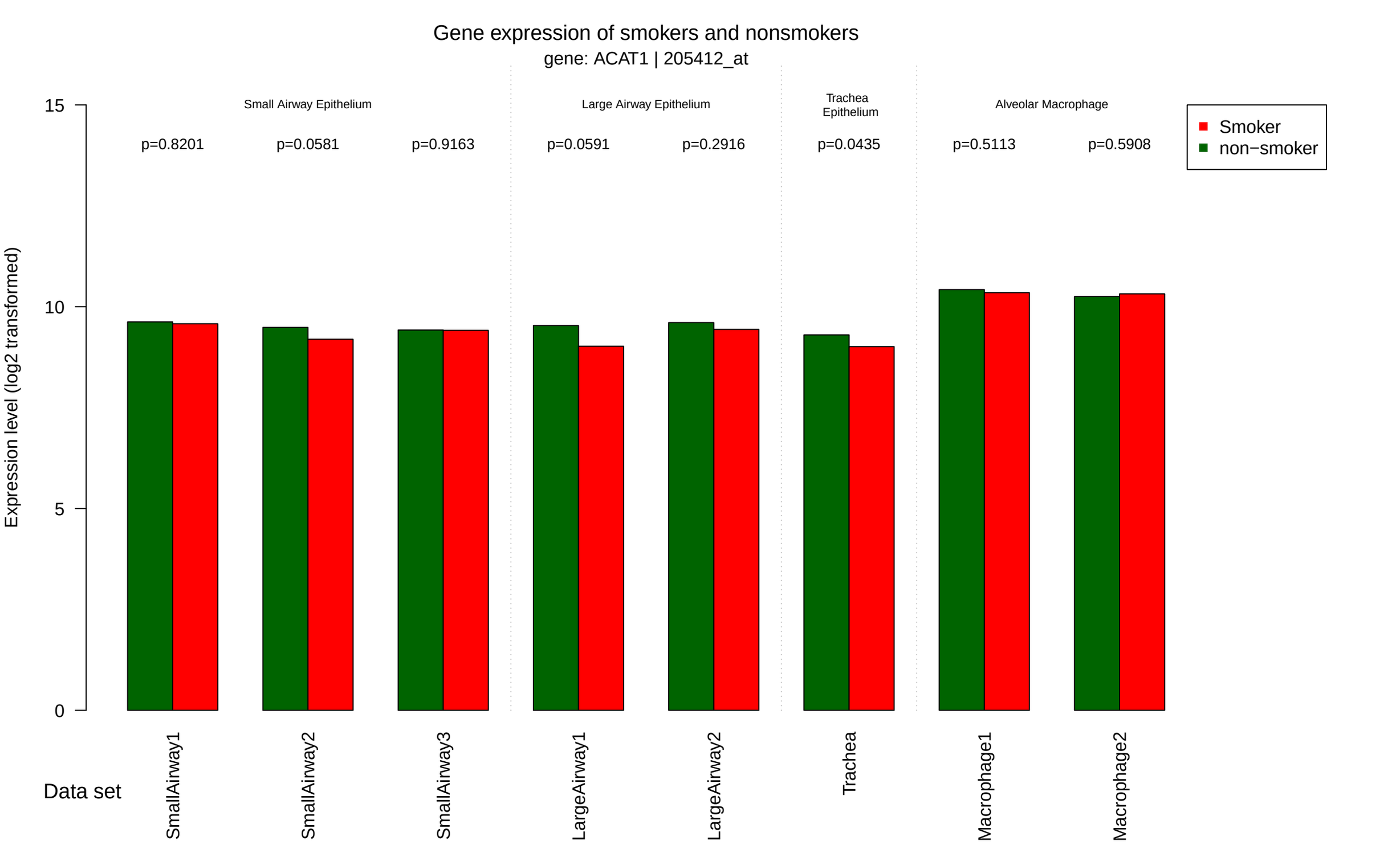
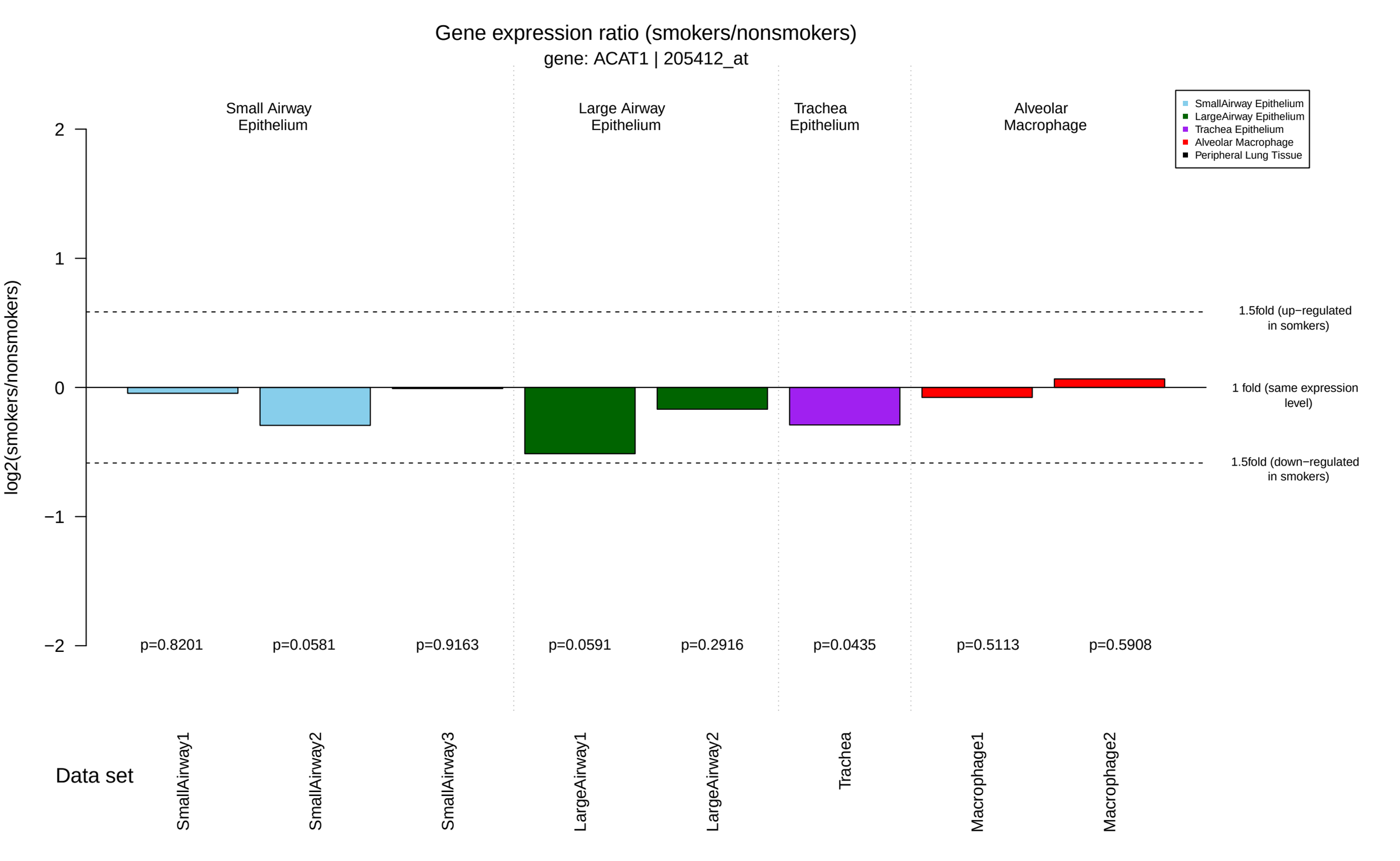
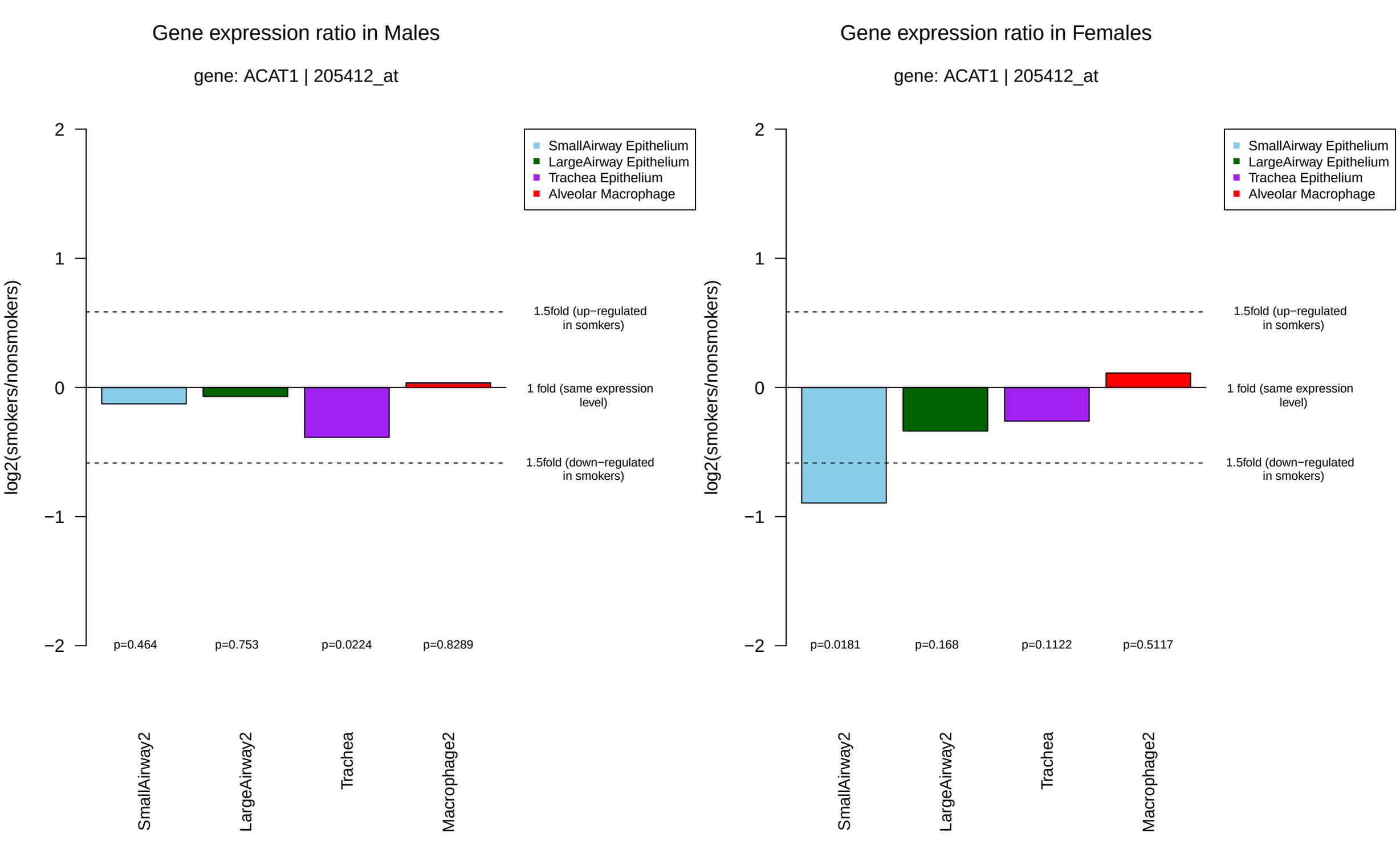
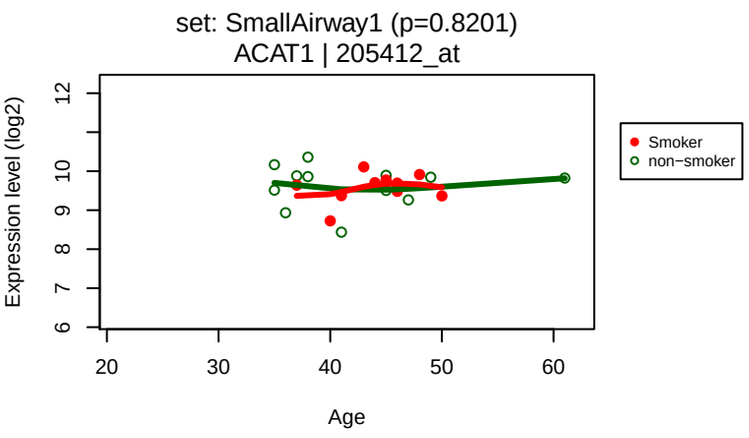
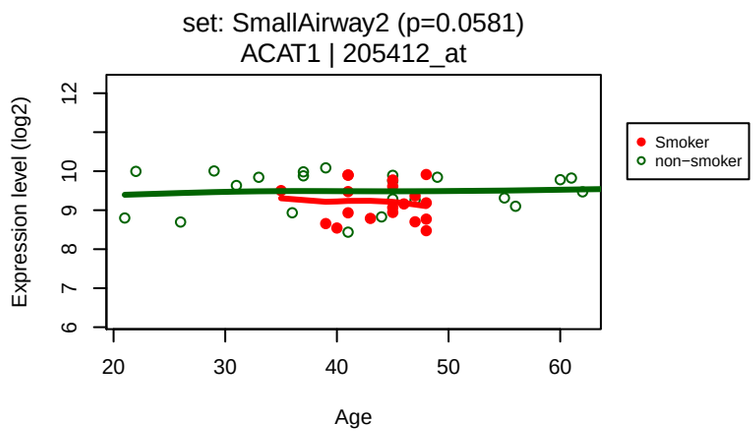
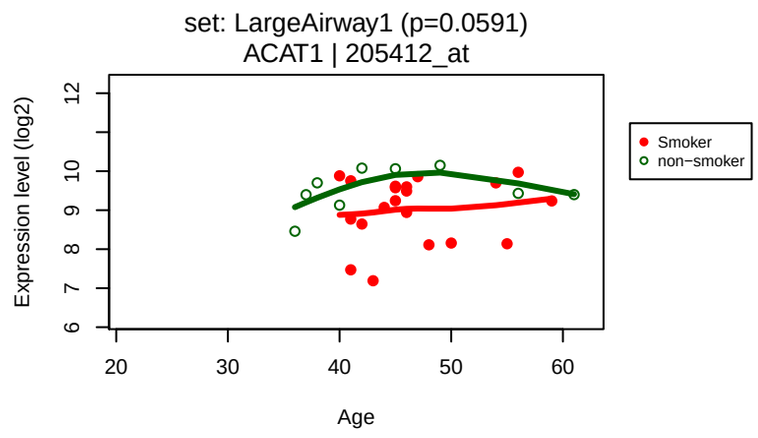

Gene expression of smokers and nonsmokers
gene: ACAT1 | 205412_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8201
p=0.0581
p=0.9163
p=0.0591
p=0.2916
p=0.0435
p=0.5113
p=0.5908
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

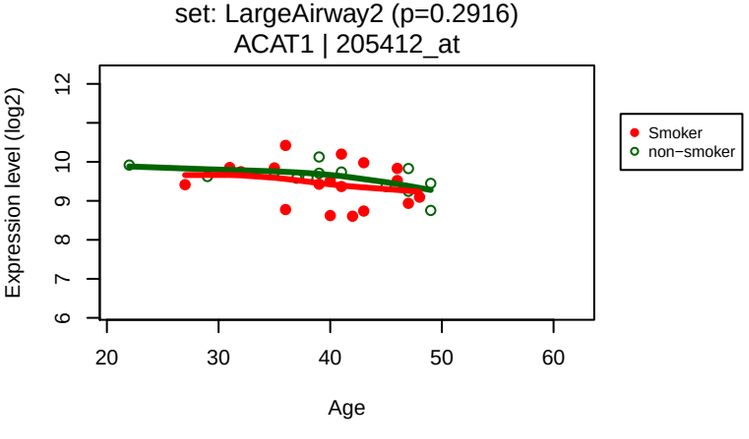

Gene expression ratio (smokers/nonsmokers)
gene: ACAT1 | 205412_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8201
p=0.0581
p=0.9163
p=0.0591
p=0.2916
p=0.0435
p=0.5113
p=0.5908
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

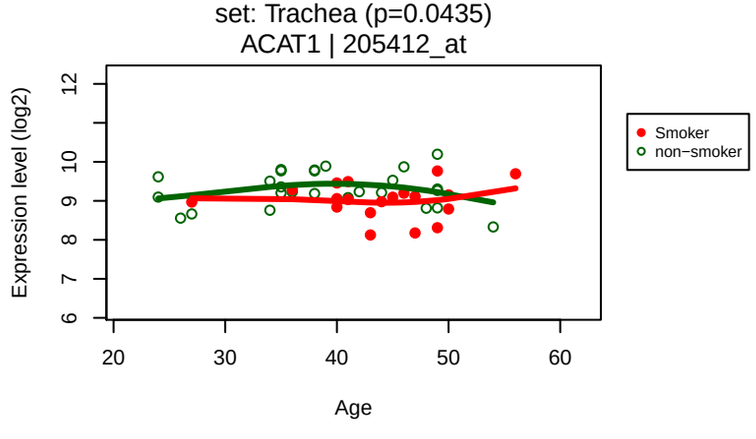

Gene expression ratio in Males
gene: ACAT1 | 205412_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.464
p=0.753
p=0.0224
p=0.8289
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ACAT1 | 205412_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0181
p=0.168
p=0.1122
p=0.5117
SmallAirway2
LargeAirway2
Trachea
Macrophage2

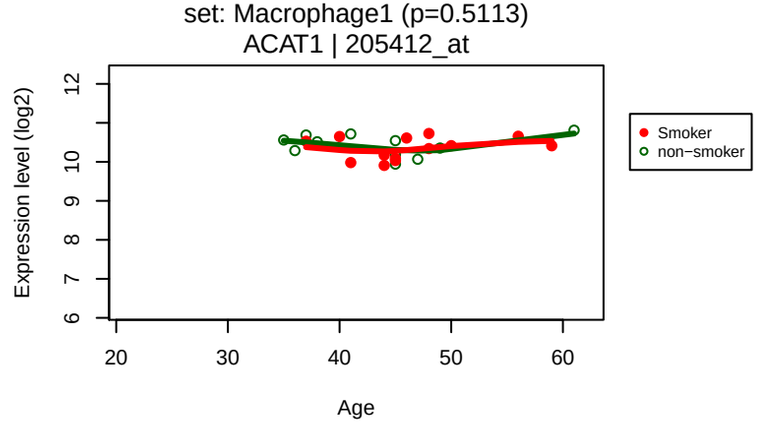

set: SmallAirway1 (p=0.8201)
ACAT1 | 205412_at
Expression level (log2)
Age
Smoker
non−smoker

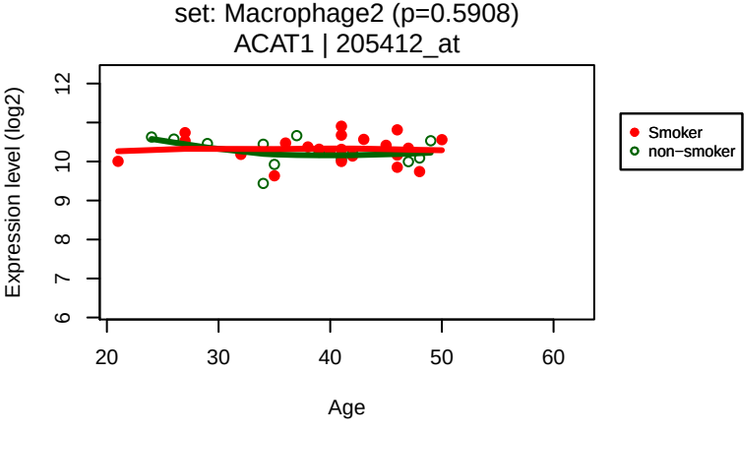

set: SmallAirway2 (p=0.0581)
ACAT1 | 205412_at
Expression level (log2)
Age
Smoker
non−smoker

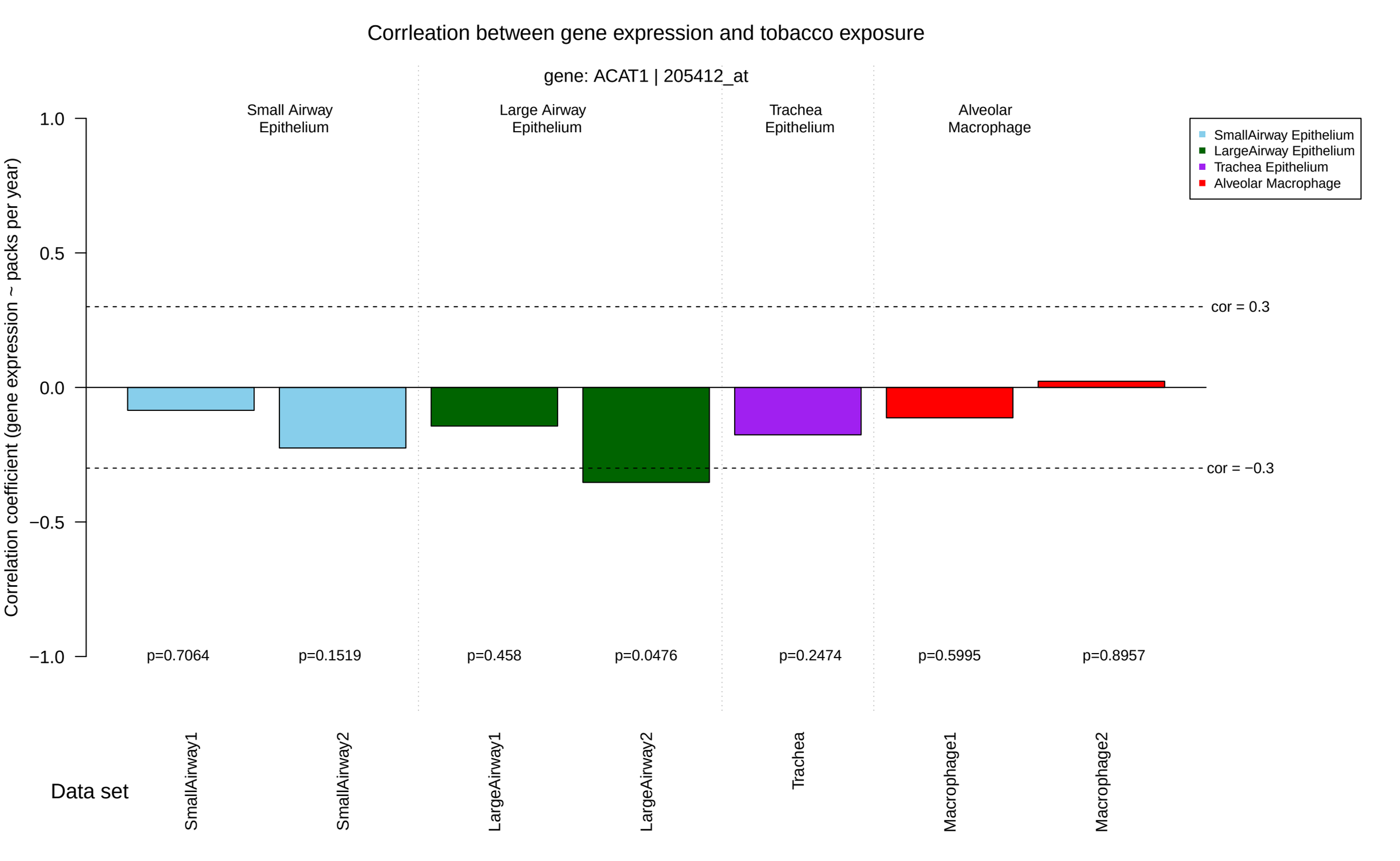

set: LargeAirway1 (p=0.0591)
ACAT1 | 205412_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2916)
ACAT1 | 205412_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0435)
ACAT1 | 205412_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5113)
ACAT1 | 205412_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5908)
ACAT1 | 205412_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ACAT1 | 205412_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7064
p=0.1519
p=0.458
p=0.0476
p=0.2474
p=0.5995
p=0.8957
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2